

GHANA MARITIME DEFENSE COMMAND (GMDC)

COMPANY OVERVIEW

Organization Profile

Attribute	Details
Organization Name	Ghana Maritime Defense Command (GMDC)
Organization Type	Fictional Government Maritime-Defense Organization
Location	Accra, Ghana
Primary Function	Maritime monitoring, coordination, communications, administration, logistics, and information security
Technology Dependence	High-core administrative and operational processes depend on information systems

Introduction

Ghana Maritime Defense Command (GMDC) is a fictional maritime-defense organization developed for this cybersecurity project. The organization is designed to coordinate maritime monitoring, operations, communications, administration, logistics, and information technology activities.

GMDC relies on interconnected computer systems, network infrastructure, applications, databases, servers, and employee workstations to support its daily operations. These systems contain important organizational and operational information, making cybersecurity an essential part of protecting the organization's confidentiality, integrity, and availability.

This document provides an overview of GMDC, including its organizational structure, departments, personnel, technological environment, and cybersecurity objectives.

Mission and Purpose

GMDC's mission is to support maritime security through effective monitoring, communication, coordination, information management, logistics, and administrative operations. The organization uses interconnected information systems to enable authorized personnel to access information and perform their assigned responsibilities.

Core Organizational Activities

- Maritime monitoring and surveillance support
- Coordination of maritime operations
- Internal and external organizational communications
- Personnel administration and records management
- Financial administration
- Logistics and resource management
- Information technology and cybersecurity management
- Document and data management

Organizational Departments

To manage its complex operations, GMDC is divided into specialized departments led by the following leadership structure

Department	Primary Responsibility
Command	Leadership, decision-making, oversight, and organizational coordination.
Operations	Maritime operations coordination and operational information management.

IT & Cybersecurity	Network, systems, cybersecurity, user support, and technical administration.
Human Resources	Personnel records, employee administration, and HR processes.
Finance	Financial records, budgeting, and financial administration.
Logistics	Resource coordination, inventory, procurement, and logistics records.
Communications	Organizational communications, public information, and email services.

Personnel and Endpoint Structure

The organization operates highly restricted endpoints. Below is the device allocation for the primary operational team.

Device	Personnel	Position	Department
PC-01	Micheal Afrifa	Director-General	Command
PC-02	Shalom Mantey Addison	Operations Manager	Operations
PC-03	Sherif Dokurugu Haruna	Maritime Operations Officer	Operations
PC-04	Stella Boateng	IT Manager	IT & Cybersecurity
PC-05	Rita Adjei	Cybersecurity Analyst	IT & Cybersecurity
PC-06	Tetteh Robert	Network Administrator	IT & Cybersecurity
PC-07	Mohammed Abdul	HR Manager	Human Resources

PC-08	Zipporah Frimpong	Finance Officer	Finance
PC-09	Samuel Asante	Logistics Officer	Logistics
PC-10	Otis Owusu	Communication Officer	Communications

Technology Environment

The GMDC environment contains a layered network architecture. Public-facing services are separated from the internal network through a DMZ, while employee endpoints and internal servers are connected through departmental and core network infrastructure.

- **Network Infrastructure:** Internet connection, firewall, router, core switch, and departmental switches.
- **DMZ Services:** Public web server and email gateway/server.
- **Employee Endpoints:** Ten PCs representing the ten personnel listed above.
- **Operational Systems:** Command Management System and Maritime Information System.
- **Business Systems:** Personnel/HR system and Financial System.
- **Data Services:** File server and backup server.
- **Security/Monitoring:** Logging, monitoring, endpoint protection, and access control functions.

High-Level Information Flow

External users reach GMDC's public-facing services through the Internet and firewall. Public services are placed in the DMZ. Authorized internal users connect through departmental switches and the core network to access systems appropriate to their roles. Critical internal applications and databases are not intended to be directly exposed to the public Internet.

Layer	Main Components
External	Internet/External Users
Security Boundary	Firewall
DMZ	Public Web Server, Email Gateway
Network Core	Router, Core Switch
Departments	Command/Operations, IT & Cybersecurity, HR/Finance/Admin
Endpoints	PC-01 through PC-10
Critical Services	Command Management, Maritime Information, Personnel, Finance, File and Backup Servers

Cybersecurity Objectives

- **Protect confidentiality:** Prevent unauthorized disclosure of sensitive organizational, personnel, financial, and operational information.
- **Protect integrity:** Prevent unauthorized modification or destruction of important records and configurations.
- **Maintain availability:** Keep essential systems and services available to authorized users.
- **Control access:** Ensure users receive only the permissions required for their roles.
- **Detect threats:** Use logging, monitoring, and security controls to identify suspicious activity.
- **Support recovery:** Maintain protected backups and recovery procedures for important systems and data.

Cybersecurity Scope for Week 1

- Define the fictional organization and its operational environment.
- Identify critical technical and information assets.
- Create a structured asset inventory and assign criticality.
- Develop a high-level network diagram showing systems and interactions.
- Identify possible entry points and attack surfaces.
- Develop a basic threat model using the STRIDE framework.

Conclusion

The Ghana Maritime Defense Command (GMDC) provides a fictional but structured environment for the Week 1 cybersecurity assessment. The organization contains ten personnel represented by ten endpoint PCs, several departments, public-facing services, internal applications, databases, servers, and network infrastructure. This overview establishes the organizational context required to perform the subsequent asset inventory, network architecture, attack-surface identification, and STRIDE threat-modeling activities.